Indian Institute of Technology Dharwad

Department of Computer Science & Enginnering

Lab Assignment 4

Pipelined Processor Simulator for ToyRICS

CS301 ---- Computer Architecture

Rhishikesh Sandip Tayade

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Benchmark Results

Program	Cycles	No. of OF stages Stalled	Wrong Path Instruction
Even/Odd	16	6	0
Palindrome	107	47	7
Prime	53	15	5
Fibonacci	120	22	160
Descending Sort	459	90	88

Notes :

Data Hazards*: Number of instructions that encountered a RAW (Read After Write) dependency.*

Control Hazards*: Number of taken branch or jump instructions that caused a pipeline flush.*

Observations: *One can observe that the number of cycles in a pipelined processor are almost half as many of those which we found in for a non-pipelined processor. Theoretically, we expect this value to be 1/5 th of the non-pipelined number of cycles. This disrepancy can be assigned to the various kinds of data hazards that are faced while execution (and hence the implementation of NOPs which stall the cycle and hence increase their number).*